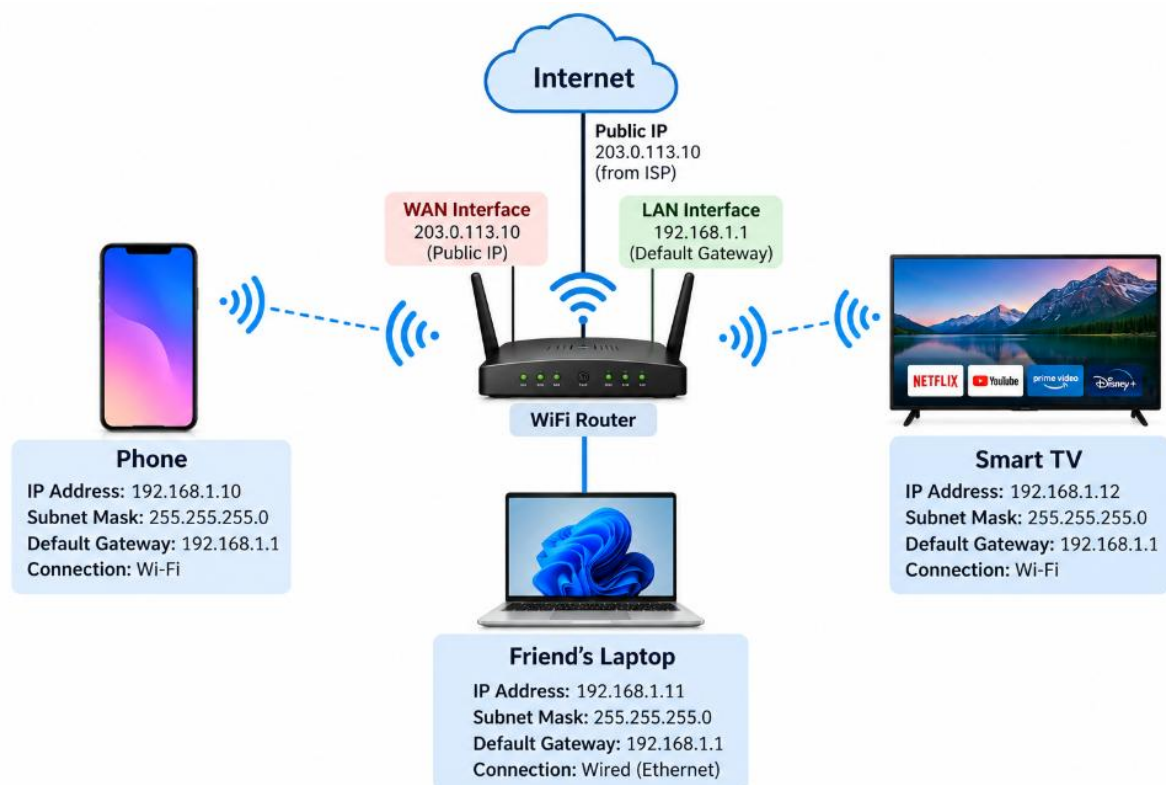


IP Routing & Routing Protocols

Task – 1 : Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

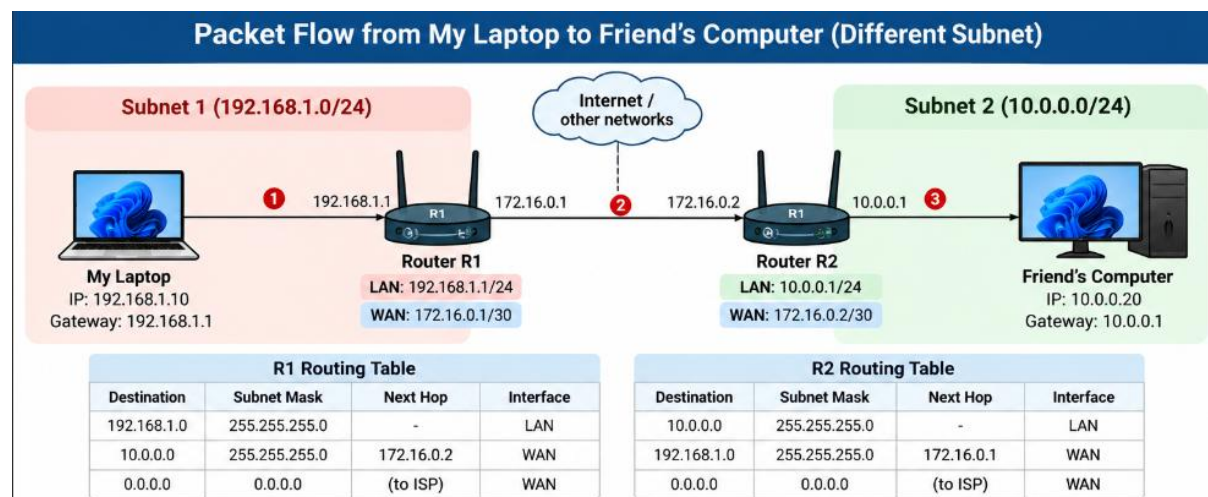
Solution : **Network Diagram with IP Addresses**

A router connects different networks and uses its routing table to forward packets between them.



Task – 2 : Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made. Hint : Use made-up IP addresses and show the routing table entries that would be checked at each hop.

Solution : **Manual IP Routing**



How it works :

Step 1: Laptop Checks the Destination

The laptop wants to send a packet to:

Destination = 10.0.0.20

The laptop's own network is : 192.168.1.0/24

Since 10.0.0.20 is **not on the local network**, the laptop sends the packet to its default gateway : 192.168.1.1

Packet : 192.168.1.10 → 10.0.0.20

Step 2: R1 Checks Its Routing Table

R1 receives the packet and checks its routing table:

Destination	Subnet Mask	Next Hop	Interface
192.168.1.0	255.255.255.0	Direct	LAN
10.0.0.0	255.255.255.0	172.16.0.2	WAN
0.0.0.0	0.0.0.0	ISP	WAN

R1 finds a route for : 10.0.0.0/24

The next hop is : 172.16.0.2

So R1 forwards the packet to **R2**.

Step 3: R2 Checks Its Routing Table

R2 receives the packet and checks its routing table:

Destination	Subnet Mask	Next Hop	Interface
10.0.0.0	255.255.255.0	Direct	LAN
192.168.1.0	255.255.255.0	172.16.0.1	WAN
0.0.0.0	0.0.0.0	ISP	WAN

R2 sees that 10.0.0.20 belongs to its directly connected network:

10.0.0.0/24

So R2 forwards the packet through its LAN interface to:

10.0.0.20

Task – 3 : Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

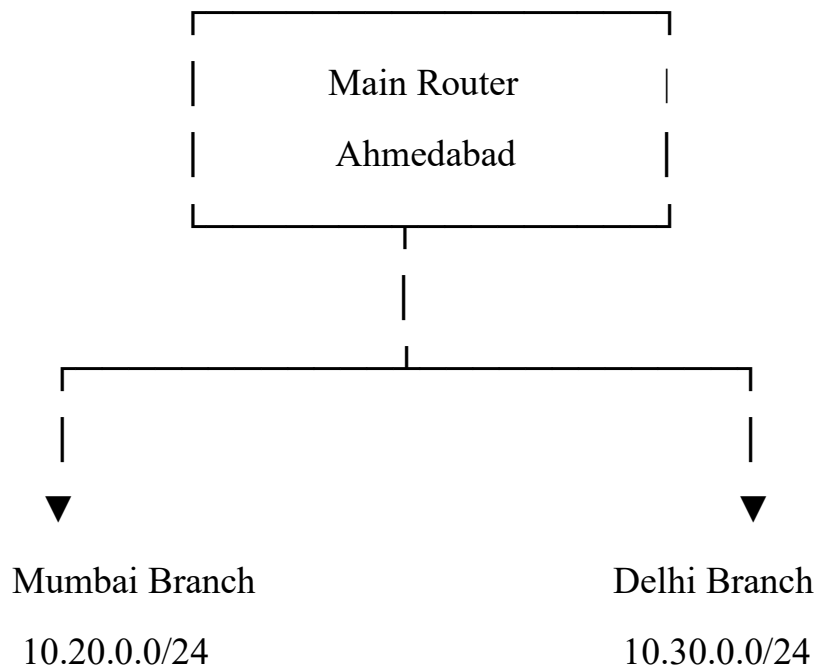
Solution : Zomato-Style Delivery Network Routing Table

Suppose a fictional food delivery company has three branches:

- Ahmedabad
- Mumbai
- Delhi

Each branch has its own subnet.

Network Diagram



Branch Networks

Branch	Network Address	Subnet Mask
Ahmedabad	10.10.0.0/24	255.255.255.0
Mumbai	10.20.0.0/24	255.255.255.0
Delhi	10.30.0.0/24	255.255.255.0

Assume the Ahmedabad router has:

G0/0 = 10.10.0.1

G0/1 = 10.0.0.1

G0/2 = 10.0.0.5

And the next-hop routers are:

Mumbai Router = 10.0.0.2

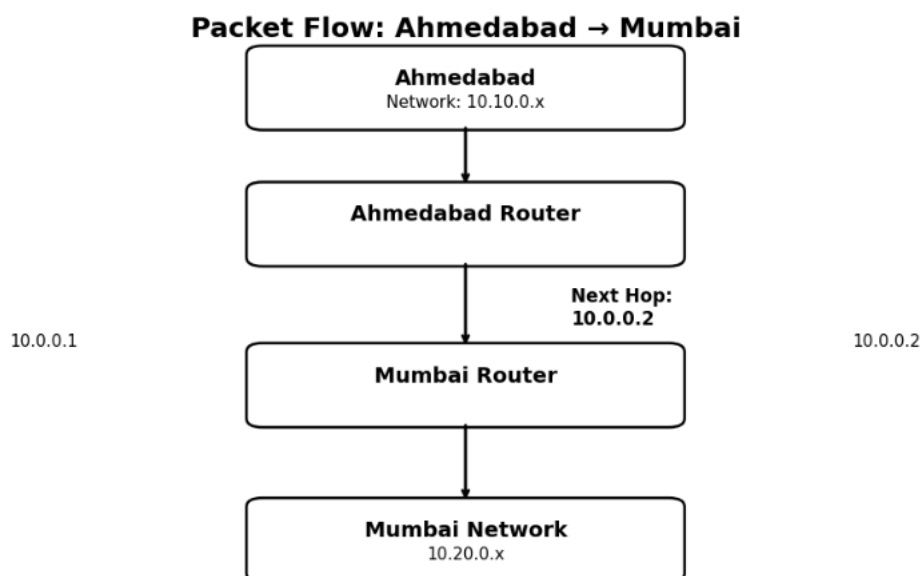
Delhi Router = 10.0.0.6

Routing Table

Destination Network	Subnet Mask	Next Hop	Interface
10.10.0.0	255.255.255.0	Direct	G0/0
10.20.0.0	255.255.255.0	10.0.0.2	G0/1
10.30.0.0	255.255.255.0	10.0.0.6	G0/2

Example

If an Ahmedabad restaurant needs to communicate with a Mumbai server:



The router looks at the **destination IP address**, finds the matching network in its routing table, and forwards the packet through the specified next hop/interface.

Task – 4 : Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

Solution : **Static Routing vs Dynamic Routing**

Comparison

Feature	Static Routing	Dynamic Routing
Configuration	Manually configured	Routes learned automatically
Example	Administrator adds routes	RIP, OSPF
Changes	Administrator must update routes	Protocol can learn network changes

Static Routing

Advantages

1. Simple for small networks

For a small restaurant or office with only a few networks, manually configured routes can be sufficient.

2. No routing protocol overhead

Routers do not need to exchange routing information continuously.

3. More predictable

The administrator explicitly defines which path should be used.

Disadvantages

1. Difficult to maintain in large networks

A large food delivery company with many branches would require many manual routes.

2. Does not automatically adapt to failures

If a configured path fails, another route does not automatically have to be discovered.

3. Configuration errors are possible

Incorrect manual routes can cause packets to be dropped or sent to the wrong destination.

Dynamic Routing

Examples including **RIP and OSPF**.

Advantages

1. Automatically learns routes

Routers exchange routing information and learn about available networks.

2. Can adapt to network changes

If a route becomes unavailable, the routing protocol can calculate another available route.

3. Suitable for larger networks

A service with many branches and data centers can reduce the amount of manual route configuration.

Disadvantages

1. More complex

The network administrator needs to understand and configure the routing protocol.

2. Uses network and router resources

Routers exchange routing information and maintain routing tables.

3. Configuration can be more difficult to troubleshoot

Incorrect protocol settings, areas, metrics, or network statements can cause routing problems.

Real-World Example

Imagine a food delivery company has servers in **Ahmedabad, Mumbai, Delhi, and Bengaluru**.

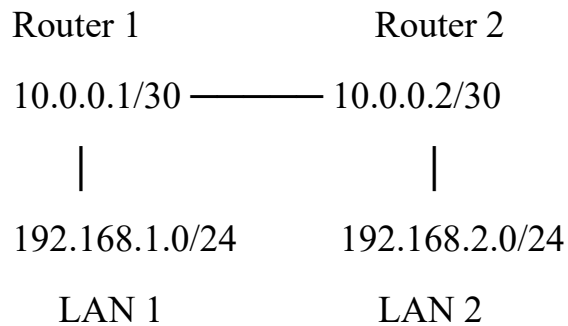
With static routing, administrators would manually configure routes between these networks.

With dynamic routing such as **OSPF**, routers can exchange routing information and learn available paths automatically. If one link becomes unavailable, OSPF can recalculate the available route.

Task – 5 : Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does. Hint : Paste the AI-generated config and annotate each part with a short description.

Solution : **Sample OSPF Configuration**

Network



Router 1

```
enable
configure terminal

router ospf 1
router-id 1.1.1.1
network 10.0.0.0 0.0.0.3 area 0
network 192.168.1.0 0.0.0.255 area 0

end
write memory
```

Router 2

```
enable
```

configure terminal

router ospf 1

router-id 2.2.2.2

network 10.0.0.0 0.0.0.3 area 0

network 192.168.2.0 0.0.0.255 area 0

end

write memory

Explanation

Command	Purpose
enable	Enters privileged mode
configure terminal	Enters configuration mode
router ospf 1	Starts OSPF process
router-id	Gives the router a unique OSPF ID
network ... area 0	Enables OSPF and advertises the network
area 0	OSPF backbone area
end	Exits configuration mode
write memory	Saves the configuration

Verify OSPF

show ip ospf neighbor

show ip route

OSPF-learned routes appear with **O**.

Example:

O 192.168.2.0/24

This means Router 1 learned 192.168.2.0/24 through OSPF.